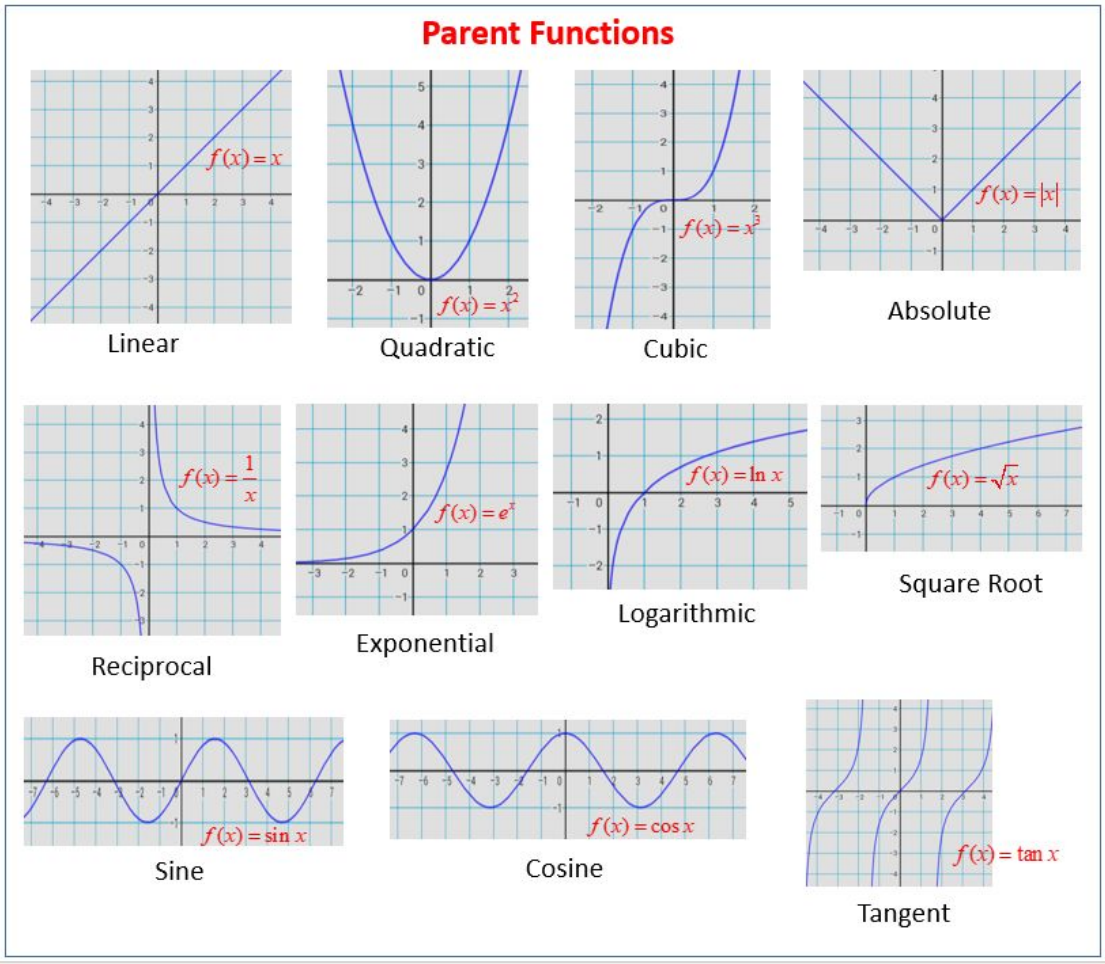
Transformations are changes to the original or parent function. It tweaks the parent function so it is different but also still resembles the same traits as the parent function. The following are some examples of parent functions.



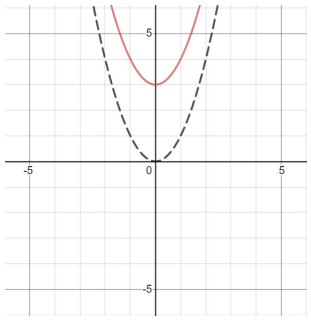
A person would tweak these functions to be different from the parent function but still represent them. For example, the parent function of a quadratic is $f(x) = x^2$. Usually during math you wouldn't encounter the parent function but you would receive variations or transformations of the parent function like $-x^2 + 4x + 5$. This equation would still be a parabola and have a similar shape to the parent function but it would be different. It received transformations to make it different.

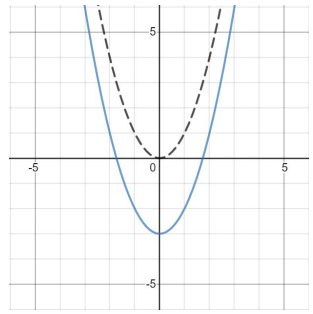
There are 4 different kinds of transformations: translation, rotation, dilations, and reflection.

Transformations have rules that must be followed. These rules are the easier ones.
(**Note**: $f(x)$ = the function)

$f(x) + b$
shifts the function b units upward.
Ex: $x^2 + 3$

$f(x) - b$
shifts the function b units downward.
Ex: $x^2 - 3$

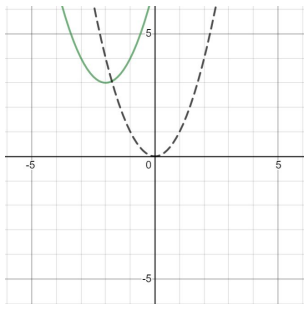




$f(x+b)$

shifts the function b units to the left.

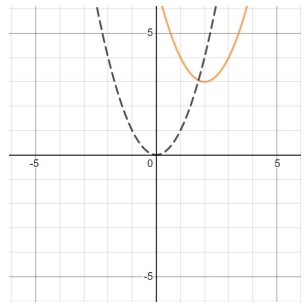
Ex: $(x+2)^2+3$



$f(x-b)$

shifts the function b units to the right.

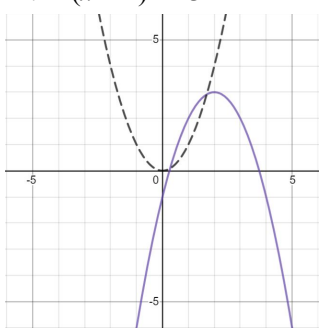
Ex: $(x-2)^2+3$



$-f(x)$

reflects the function in the x-axis.

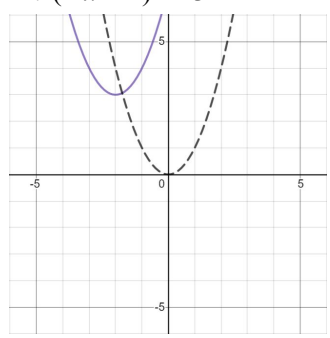
Ex: $-(x-2)^2+3$



$f(-x)$

reflects the function in the y-axis.

Ex: $(-x-2)^2+3$



$a\,f(x)$

Creates vertical stretch or compression

If $a>1$, then graph stretches vertically

If $0<a<1$, then graph is compressed vertically

Ex: $2(x)^2$

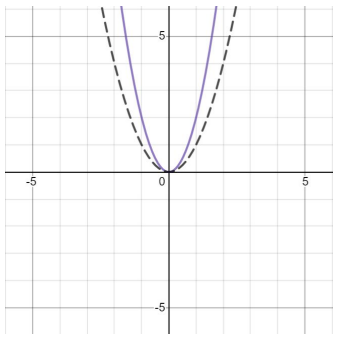
$f(bx)$

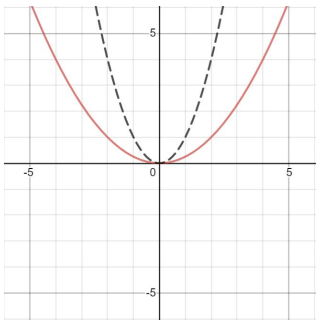
Creates horizontal stretch or compression

If $b>1$, then graph comprises horizontally

If $0<b<1$, then graph stretches horizontally

Ex: $(\frac{1}{2}x)^2$



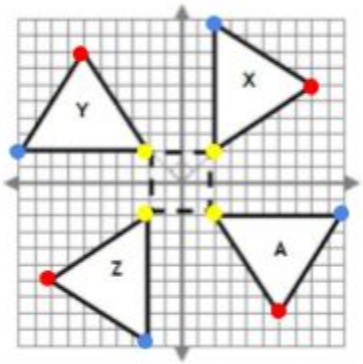


Rotations

Usually when given a problem, they will ask you to rotate counterclockwise by either 90°, 180°, or 270°.

To rotate a shape or function, take the points that you already know and rotate it according to these following rules.

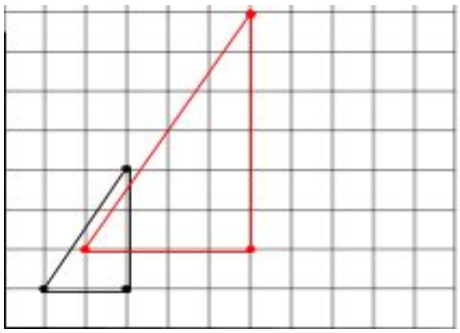
Type of rotation	Original point	Rotated point
counterclockwise 90°	(x, y)	(-y, x)
counterclockwise 180°	(x, y)	(-x, -y)
counterclockwise 270°	(x, y)	(y, -x)



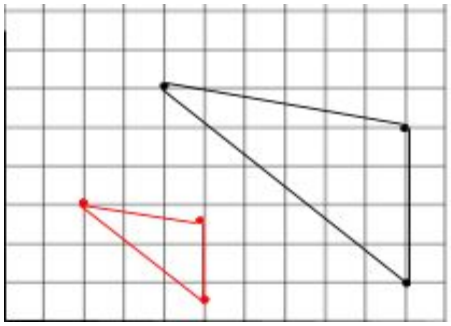
Dilations

A dilation is a transformation that produces the same shape but a different size. It must have a center of dilation. This is a fixed point that the shape would dilate around. Most likely the center of dilation would be around the origin or (0, 0).

An Enlargement creates a bigger image. This is created by multiplying each coordinate by factor.



A Reduction creates a smaller image with the same process.



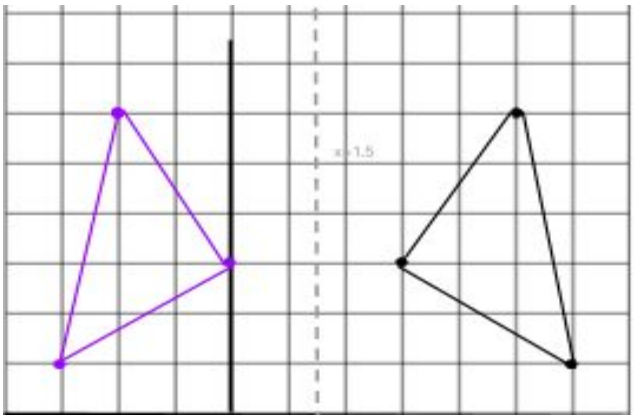
Reflection

Some of these rules were already talked about earlier. These are the only rules you need to know for reflection.

Coordinate Rules for Reflection

Reflection over	
x-axis	$(x,y)\rightarrow(x,-y)$
y-axis	$(x,y)\rightarrow(-x,y)$
y = x	$(x,y)\rightarrow(y,x)$
y = -x	$(x,y)\rightarrow(-y,-x)$

Some reflections are not over the x or y-axis but over a x or y line (Ex: x=3). For these we need to find the difference from each point to the reflections line to reflect correctly.



For this example, the black triangle was reflected over x=1.5 to get the purple triangle. When doing this by hand, each distance from each point to the reflection line needs to be equal to the corresponding triangle point. For the black triangle there were 3 different distances from the reflection line.

Writing Transformations

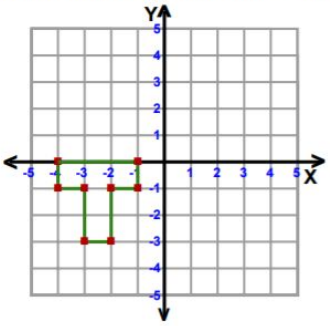
(Note: Make sure to write transformations in the following order because sometimes the order can impact the outcome of a function.)

1. Start with parentheses (look for possible **horizontal** shift)
2. Deal with multiplication (**stretching** or **compressing**)
3. Deal with negation (**reflection**)
4. Deal with addition/subtraction (**vertical** shift)

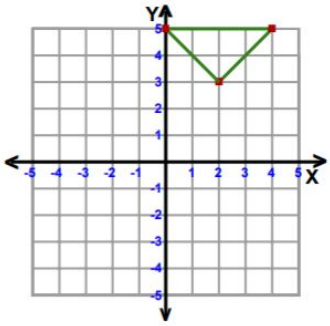
Playing around with Desmos can personally help us understand how graphs move visually. When you finish with Desmos, here are some practice problems.

Practice problems

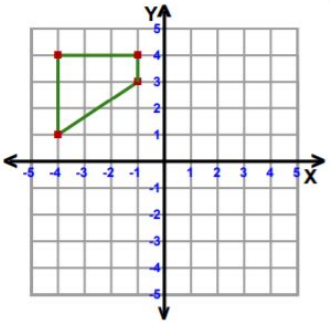
1) Translation: 2 right and 3 up



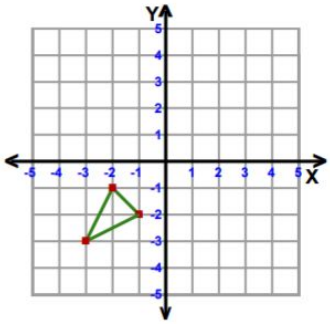
2) Reflection: Across the x-axis



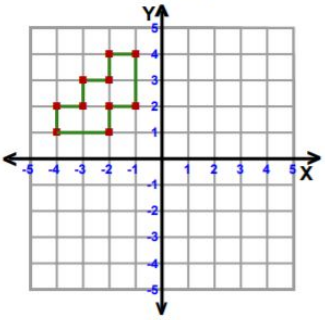
3) Rotation: 180° about the origin



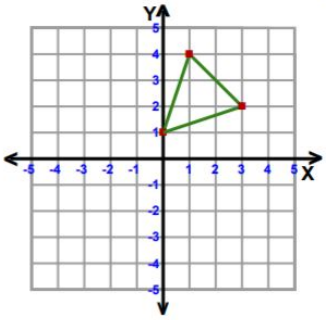
4) Rotation: 90° ccw about the origin



5) Translation: 4 right and 5 down



6) Reflection: Across the line y = 2



https://www.math-aids.com/cgi/pdf_viewer_4.cgi?script_name=transformation_combined.pl&p_trans=1&p_rotate=1&p_reflect=1&triangles=1&polygons=1&boxes=1&language=0&memo=&answer=1&x=145&y=25